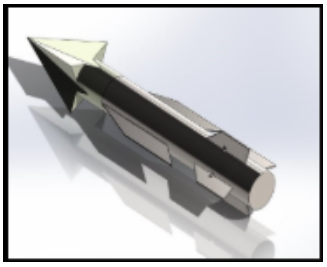
2026 North Carolina A&T Hypersonic Senior Design
Executive Summary and Information



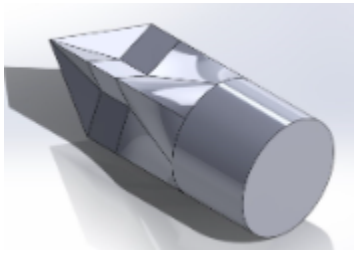
Project Overview

This project presents the design and preliminary validation of an unmanned hypersonic projectile capable of stable and controlled flight within the Mach 5 to Mach 8 regime. Developed as part of the Undergraduate Hypersonic Flight Design Competition, the system is constrained to a maximum length of 1.0 m, a diameter of 0.2 m, and a total mass of 45 kg. The design emphasizes aerodynamic efficiency, structural integrity, and thermal survivability under hypersonic conditions.

Engineering Innovation

The vehicle incorporates a novel four-point caret-wing shock generator, enabling controlled oblique shock formation and efficient airflow compression. This design leverages favorable principles to enhance lift generation and improve lift-to-drag ratio, which is critical for unpowered hypersonic glide performance.

Key features include:

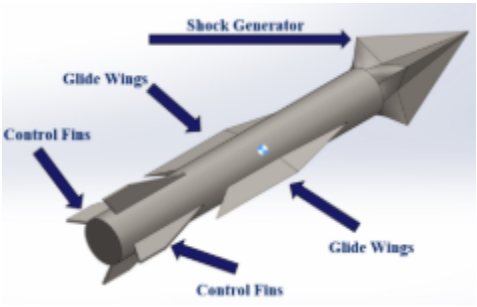


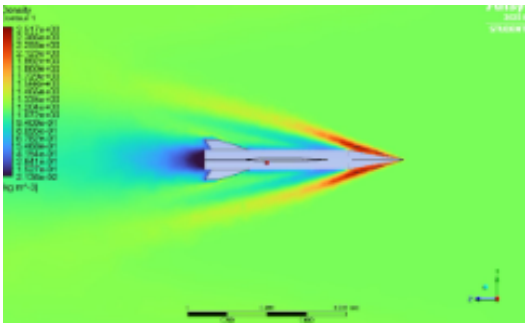
- Caret-wing geometry for shock-controlled lift generation
- Low-aspect-ratio wings to enhance glide efficiency
- Highly swept fins for aerodynamic stability and control
- Shock launch system to enable high-acceleration deployment (~30,000 g)

Performance Highlights

- Operational Speed: Mach 5–8
- Static Margin: ~21% (stable and controllable flight)
- Maneuver Capability: ≥ 15 g at 9 km altitude
- Thermal Constraint: Internal payload ≤ 74°C
- Design Focus: Maximize lift-to-drag ratio for extended glide range

Analytical and simulation-based results indicate that the design achieves sufficient aerodynamic efficiency and structural robustness to meet these performance targets.



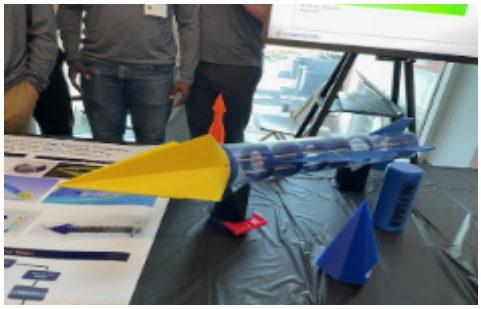


Design & Analysis Approach

The project integrates a multidisciplinary engineering approach combining:

- Aerodynamic analysis using compressible flow and oblique shock relations
- CAD modeling for geometry development and integration
- CFD and FEA simulations (ANSYS, SolidWorks) for aerodynamic and structural validation
- Material optimization using titanium alloys, tungsten, and thermal protection systems

A full-scale 3D-printed mockup was developed to validate manufacturability, subsystem integration, and assembly feasibility.



Testing & Validation Strategy

To ensure real-world applicability, the team developed a validation approach combining simulation, controlled testing, and subsystem-level experimentation:

- Wind tunnel testing (in collaboration with North Carolina State University) to evaluate aerodynamic performance, including lift, drag, and stability characteristics
- Instrumentation system (IMU, GPS, Kalman filtering) for real-time data acquisition and system validation
- Sabot separation testing using a controlled low-cost launch platform (potato cannon) to evaluate mechanical separation behavior and ensure clean release dynamics

With a full-scale launch out of the scope of the project timeline, the start of a sabot design gives a springboard for the next team to further develop the delivery of the projectile. Looking into aspects such as separation time, stability in release and potential interferences.

Industry Relevance

This project addresses key challenges in modern hypersonic systems, including:

- **Aerodynamic efficiency in high-speed regimes**
- **Thermal protection under extreme heating**
- **Structural survivability during high-G launch conditions**
- **Integration of simulation and experimental validation**

The methodologies and design strategies developed are directly applicable to aerospace, defense, and advanced vehicle systems.

Conclusion

This work demonstrates a feasible, manufacturable, and high-performing hypersonic glide vehicle design. By combining advanced aerodynamic concepts with robust structural and thermal solutions, the project establishes a strong foundation for future experimental validation and potential real-world application.